Addressing Extreme Class Imbalance in Dermoscopic Skin Lesion Classification via Class-Balanced Loss and Two-Stage Fine-Tuning

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Abstract

Automated skin lesion classification using dermoscopic images has the potential to support early detection of malignant skin cancers; however, model performance is often severely degraded by extreme class imbalance and subtle inter-class visual similarity. While modern convolutional neural networks achieve high overall accuracy, rare lesion types of significant clinical importance are frequently misclassified or ignored entirely. In this work, we develop a multiclass skin lesion classification framework that explicitly addresses imbalance-induced performance collapse. Using the HAM10000 dataset, we combine class-balanced cross-entropy loss, ImageNet-based transfer learning, and a two-stage fine-tuning strategy to stabilize training and improve minority-class discrimination. Our results demonstrate substantial gains in underrepresented categories, including dermatofibroma and actinic keratoses, while preserving strong performance on majority classes. This study highlights the critical role of loss design and training strategy in deploying deep learning models for real-world medical imaging tasks.

Keywords: Skin cancer classification; class imbalance; transfer learning; dermoscopy; deep learning

1 Introduction

Skin cancer is among the most prevalent forms of cancer worldwide, and early diagnosis remains a key determinant of patient outcomes. Dermoscopic imaging has become an essential diagnostic tool, enabling clinicians to visualize lesion structures beyond what is visible to the naked eye. Despite its utility, accurate interpretation of dermoscopic images requires substantial expertise, and diagnostic agreement between dermatologists can vary considerably.

Recent advances in deep learning have led to significant progress in automated skin lesion classification. Convolutional neural networks (CNNs) are capable of learning hierarchical visual representations and have demonstrated performance comparable to, or exceeding, traditional machine learning approaches. However, in clinical datasets, these gains are often unevenly distributed across classes. Rare but clinically critical lesion types remain persistently underdetected, undermining the reliability of automated systems in real-world settings.

A fundamental challenge lies in the extreme class imbalance inherent to dermatological datasets. Benign lesions, particularly melanocytic nevi, dominate available data, while malignant or atypical

lesions appear infrequently. When trained with standard loss functions, deep models disproportionately optimize performance on majority classes, leading to near-zero accuracy for minority categories. High overall accuracy thus masks clinically unacceptable failure modes.

This paper investigates how targeted modifications to the training objective and optimization procedure can mitigate imbalance-driven performance collapse. Rather than proposing a novel architecture, we focus on systematically improving learning dynamics through loss reweighting, transfer learning, and fine-tuning strategy.

1.1 Contributions

The primary contributions of this work are as follows:

- A rigorous analysis of imbalance-induced failure in multiclass skin lesion classification.
- The application of class-balanced cross-entropy loss to amplify minority-class learning signals.
- A two-stage fine-tuning strategy that stabilizes optimization and adapts pretrained features to dermoscopic isive evaluation demonstrating large performance gains in rare lesion categories.

2 Related Work

Early work on automated skin lesion classification relied on traditional machine learning pipelines involving preprocessing, lesion segmentation, handcrafted feature extraction, and classical classifiers. Vidya and Karki [1] employed geodesic active contour segmentation followed by ABCD rule features, GLCM, and HOG, achieving strong performance in binary melanoma detection. While effective, such approaches depend heavily on accurate segmentation and manual feature design, limiting scalability and robustness in more complex settings.

The introduction of deep learning significantly advanced skin lesion classification. Hosny et al. [2] demonstrated that transfer learning with a pretrained AlexNet model, combined with fine-tuning and data augmentation, yields high accuracy on dermoscopic images. This work established transfer learning as a key strategy for mitigating limited data availability in medical imaging. More recent CNN-based studies have focused on architectural improvements; for example, Rahman et al. [3] proposed a NASNet-based model that further improves classification performance through network optimization and augmentation. However, these approaches primarily emphasize overall accuracy and often provide limited analysis of class-wise behavior under severe imbalance.

Beyond CNNs, transformer-based models have been explored for skin lesion classification. Xin et al. [4] introduced a vision transformer framework incorporating multiscale patch embeddings and contrastive learning, achieving strong results on the HAM10000 dataset. While effective, such models typically require increased computational resources and complex training procedures.

A systematic review by Brinker et al. [5] concluded that CNNs with transfer learning represent the dominant paradigm in skin lesion classification, particularly given the scarcity of large, publicly available datasets. The review also emphasized that high aggregate accuracy may obscure poor

performance on minority classes, highlighting the need for methods that explicitly address class imbalance and report detailed per-class results.

Motivated by these observations, the present work focuses on improving minority-class performance in multiclass dermoscopic classification through loss design and training strategy, rather than architectural complexity.

3 Dataset and Problem Setting

3.1 HAM10000 Dataset

We conduct experiments on the HAM10000 dataset [6], a widely used benchmark for dermoscopic skin lesion analysis. The dataset contains 10,015 dermoscopic images categorized into seven diagnostic classes: melanoma (MEL), melanocytic nevi (NV), basal cell carcinoma (BCC), actinic keratoses (AKIEC), benign keratosis-like lesions (BKL), dermatofibroma (DF), and vascular lesions (VASC).

The dataset exhibits severe class imbalance, as shown in Figure 1. Melanocytic nevi (NV) constitute the majority of samples, while rare classes such as dermatofibroma (DF) and vascular lesions (VASC) together account for only a small fraction of the dataset. This imbalance poses a challenging learning environment in which standard empirical risk minimization tends to prioritize majority classes and fails to adequately represent minority categories.

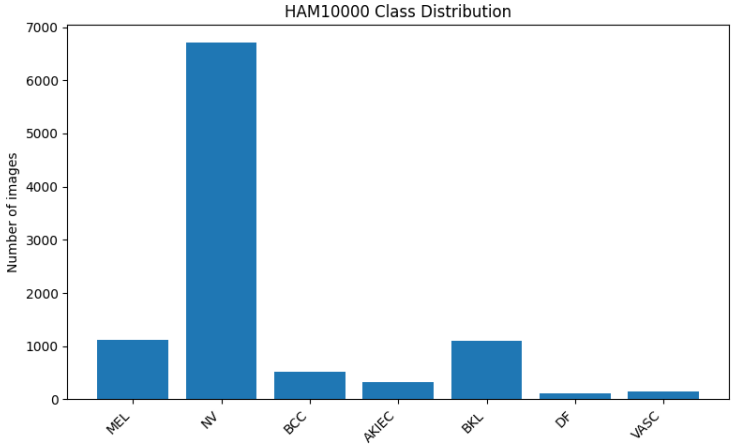

Figure 1: Class distribution of the HAM10000 dataset, illustrating the pronounced imbalance across lesion categories.

3.2 Problem Formulation

Given an input image $x \in \mathbb{R}^{224\times224\times3}$, the task is to predict a class label $y \in \{1, \ldots, 7\}$. The model learns a mapping $f_\theta(x)$ parameterized by θ, producing a probability distribution over classes. Performance is evaluated both globally and on a per-class basis.

4 Methodology

This section describes the baseline training configuration and the proposed improvements designed to address severe class imbalance in HAM10000. The final approach combines (i) ImageNet-based transfer learning for a ResNet-18 backbone, (ii) class-balanced cross-entropy loss, and (iii) a two-stage fine-tuning schedule.

4.1 Baseline Training Setup

Baseline models are trained using standard cross-entropy loss:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{C} \mathbb{I}(y = i) \log p_i, \tag{1}$$

where C is the number of classes, p_i is the predicted probability for class i, and $\mathbb{I}(\cdot)$ is the indicator function. While theoretically well-motivated, $\mathcal{L}_{CE}$ implicitly assumes class distributions that are not extremely skewed; under strong imbalance, gradients are dominated by majority classes, which can suppress learning signals for rare lesion categories.

4.2 Class-Balanced Cross-Entropy Loss

To mitigate imbalance-driven training collapse, we use class-balanced cross-entropy (CB-CE) based on the effective number of samples [7]. Let n_i denote the number of training samples in class i. The effective number is defined as

$$E_i = \frac{1 - \beta^{n_i}}{1 - \beta}, \tag{2}$$

where $\beta \in [0, 1)$ controls the strength of reweighting. The loss is then computed by applying class-dependent weights (inversely related to E_i) so that minority classes contribute proportionally more to the optimization objective than they would under standard cross-entropy.

4.3 ImageNet Pretraining with ResNet-18

We adopt transfer learning by initializing the ResNet-18 backbone with weights pretrained on ImageNet [8]. Importantly, ImageNet pretraining does *not* imply loading a larger or different architecture into our model. Instead, we instantiate the same ResNet-18 architecture and load the corresponding ImageNet-pretrained parameters for that architecture. Therefore, under standard practice, the initialization provides the *full ResNet-18 backbone* (all ResNet-18 layers) with pretrained weights, unless specific layers are intentionally removed or the weight-loading process is performed partially. In our setup, we replace the final classification layer to output seven logits, while the ResNet-18 backbone is initialized from ImageNet-pretrained weights.

4.4 Two-Stage Fine-Tuning Strategy

To improve stability under severe class imbalance and limited minority-class data, we adopt a two-stage fine-tuning strategy on an ImageNet-initialized ResNet-18 backbone. In Stage 1 (warm-up), we freeze all backbone parameters and optimize only the final fully connected (FC) classification layer to adapt the model from the original ImageNet label space to the seven HAM10000 classes. This warm-up stage is trained for 3 epochs using the Adam optimizer with learning rate 10^{-3} under the class-balanced cross-entropy objective.

In Stage 2 (fine-tuning), all layers are unfrozen and the entire network is optimized end-to-end to specialize feature representations for dermoscopic imagery. This stage is trained for 7 additional epochs using Adam with learning rate 10^{-4}, while continuing to use class-balanced cross-entropy. For both stages, we select the best model checkpoint based on validation accuracy and report final performance on the held-out test set.

4.5 Final Training Pipeline Summary

Figure 2 provides a compact overview of the final architecture and the two-stage training schedule used in this study.

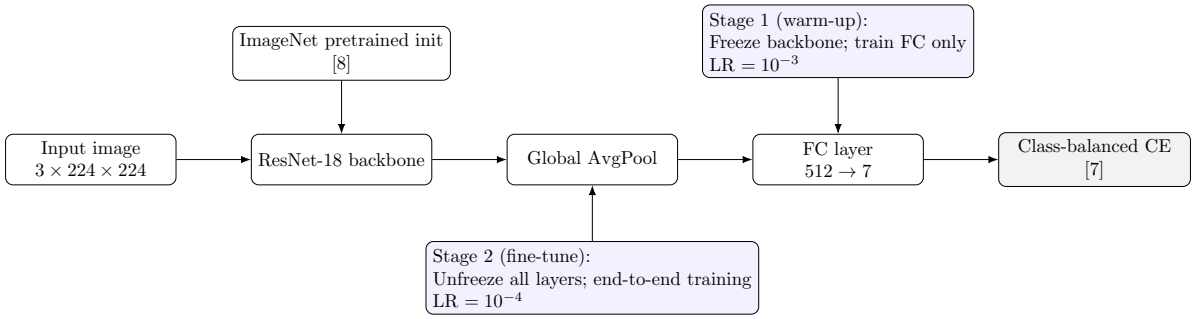

Figure 2: Overview of the final training procedure.

5 Experimental Results

This section reports quantitative performance for baseline architectures and for the proposed training improvements. In addition to overall accuracy, we emphasize class-wise behavior to highlight imbalance-driven failure modes and the effect of the proposed method on minority categories.

5.1 Baseline Model Comparison

We first evaluate several commonly used convolutional neural network architectures under a standardized baseline configuration. Table 1 reports the best validation accuracy and the corresponding test accuracy for each model. ResNet-18 achieves the highest test accuracy among the evaluated baselines and is therefore used as the primary baseline for subsequent comparisons.

Table 1: Baseline model performance (overall accuracy).

Model	Best Val Acc	Test Acc
ResNet18	0.8187	0.6885
ResNet34	0.7824	0.6772
ResNet50	0.8187	0.6779
MobileNetV2	0.7202	0.6111
EfficientNet-B0	0.7772	0.6581
EfficientNet-B1	0.7720	0.6640
VGG11	0.7358	0.6415
VGG16	0.7306	0.6601

5.2 Baseline Per-Class Performance (ResNet-18)

To understand where baseline performance breaks down, we report per-class accuracy for the best-performing baseline (ResNet-18) in Table 2. While the baseline performs strongly on the dominant class (NV), minority classes such as DF are not learned at all (0% accuracy), illustrating the severity of imbalance in this dataset.

Table 2: ResNet-18 baseline per-class accuracy on the test set.

Class	Accuracy	Correct / Total
MEL	0.497	85 / 171
NV	0.889	808 / 909
BCC	0.451	42 / 93
AKIEC	0.209	9 / 43
BKL	0.359	78 / 217
DF	0.000	0 / 44
VASC	0.543	19 / 35

5.3 Overall Performance Improvements Across Stages

We next evaluate two incremental improvements to the ResNet-18 baseline: (i) adding class-balanced cross-entropy (CB-CE), and (ii) combining CB-CE with ImageNet-pretrained initialization. Table 3 summarizes the overall test accuracy across the three stages. The final configuration yields a substantial improvement over the baseline, indicating that loss design and transfer learning jointly mitigate imbalance-driven performance collapse.

Table 3: Overall test accuracy across training stages.

Stage	Model	Test Accuracy
Baseline	ResNet18	0.6885
Improvement 1	ResNet18 + CB-CE	0.7024
Improvement 2 (Final Product)	ResNet18 (ImageNet Pretrain) + CB-CE	0.8022

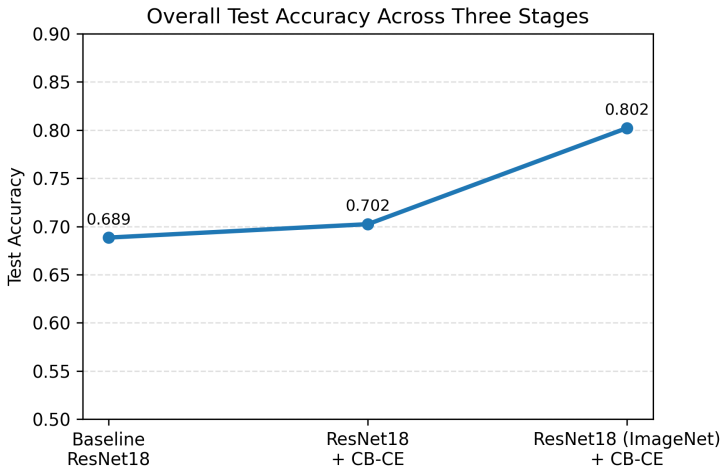

Figure 3: Overall test accuracy improvement across training stages.

5.4 Per-Class Performance Improvements

To quantify how performance changes are distributed across classes, Table 4 reports per-class accuracy for the ResNet-18 baseline and the final model. The most pronounced gains occur in minority categories (e.g., DF and AKIEC), while performance on the dominant NV class remains strong. These results indicate that the final training configuration substantially improves minority-class recognition rather than merely increasing overall accuracy through majority-class gains.

Table 4: Per-class test accuracy: baseline vs. final model.

Class	Baseline	Final	Change
MEL	0.497	0.433	$\downarrow$ 6.4%
NV	0.889	0.932	$\uparrow$ 4.3%
BCC	0.452	0.807	$\uparrow$ 35.5%
AKIEC	0.209	0.581	$\uparrow$ 37.2%
BKL	0.359	0.641	$\uparrow$ 28.1%
DF	0.000	0.659	$\uparrow$ 65.9%
VASC	0.543	0.686	$\uparrow$ 14.3%

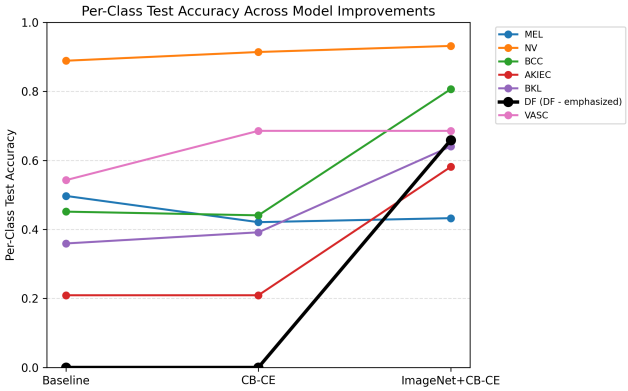

Figure 4: Per-class test accuracy across the three training stages.

5.5 Qualitative Analysis

To complement the quantitative results, we present qualitative comparisons for two clinically important minority classes: AKIEC and DF. For each class, we show representative (i) correctly classified examples, (ii) misclassified examples, and (iii) false positives (non-target classes predicted as the target class). These cases provide insight into the visual patterns that the model relies on

and the common failure modes under severe class imbalance.

5.5.1 AKIEC (Actinic Keratoses)

Correct AKIEC predictions frequently exhibit darker granular dots and keratotic (crust-like) textures. In contrast, misclassified AKIEC samples often appear smoother and lack clear dot patterns, increasing confusion with other lesion types. False positives predicted as AKIEC commonly contain dot-like or crust-like structures that visually resemble AKIEC, especially when lesion centers are ambiguous or low-contrast.

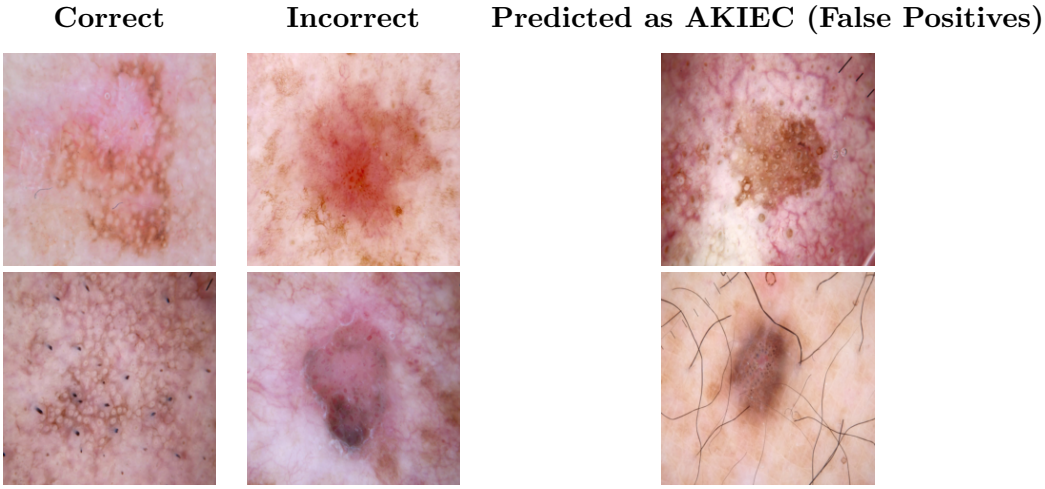

Figure 5: Qualitative results for AKIEC. Left: correctly classified AKIEC examples. Middle: AKIEC cases misclassified as other classes. Right: non-AKIEC cases predicted as AKIEC (false positives).

5.5.2 DF (Dermatofibroma)

Correct DF predictions often show a central pale/white scar-like region. Misclassified DF examples tend to exhibit reddish, irregular, or dotted appearances that resemble AKIEC, leading to frequent confusion between these two categories. False positives (non-DF predicted as DF) commonly contain circular pale centers similar to DF, suggesting that central brightness and symmetry can dominate the classifier's decision in visually ambiguous cases.

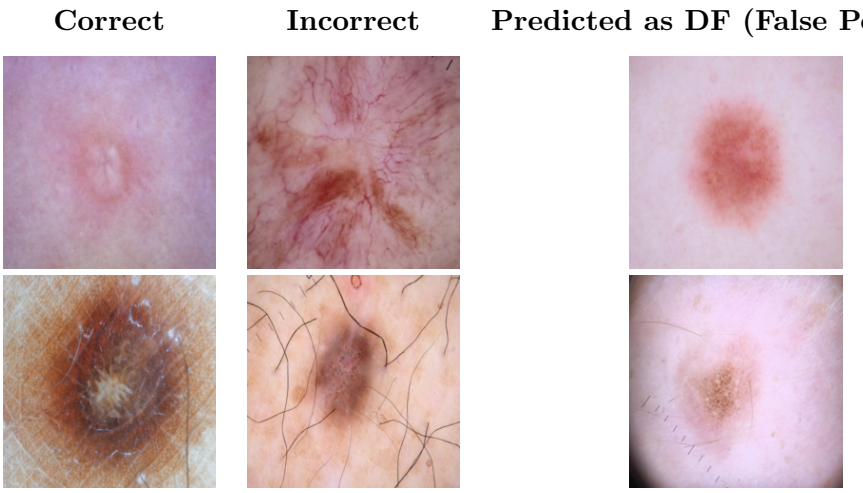

Figure 6: Qualitative results for DF. Left: correctly classified DF examples. Middle: DF cases misclassified as other classes (commonly AKIEC). Right: non-DF cases predicted as DF (false positives).

6 Discussion

The results indicate that architectural complexity alone is insufficient to address severe class imbalance in medical image classification. While deeper or more expressive networks can improve overall representational capacity, our baseline comparisons show that such models still struggle on underrepresented classes. In contrast, incorporating class-balanced cross-entropy substantially reshapes the optimization objective, allowing minority classes to contribute more effectively to training. The two-stage fine-tuning strategy further highlights the importance of training dynamics in low-data regimes, as gradually adapting the model prevents unstable updates and improves convergence.

Importantly, the observed gains are driven primarily by improvements in minority-class recognition rather than marginal increases on dominant categories. This behavior is critical in clinical settings, where rare lesion types often carry disproportionate diagnostic significance. Together with qualitative observations, these findings suggest that careful loss design and staged optimization provide a more reliable path toward robust performance in imbalanced medical imaging tasks than increasing architectural complexity alone.

7 Conclusion and Future Work

This study demonstrates that extreme class imbalance in dermoscopic image classification can be effectively mitigated through a combination of principled loss reweighting, transfer learning, and staged optimization. Starting from a standard ResNet-18 baseline, we show that incorporating class-balanced cross-entropy improves performance on underrepresented lesion categories, and that additional gains are achieved by initializing the network with ImageNet-pretrained weights followed

by a two-stage fine-tuning strategy. Importantly, these improvements are driven primarily by substantial accuracy gains in minority classes such as DF, AKIEC, and BCC, underscoring the clinical relevance of the proposed approach.

Beyond overall accuracy, per-class and qualitative analyses indicate that the final model learns more discriminative features for rare lesion types and reduces systematic failure modes observed in baseline models. These results suggest that careful optimization design can significantly improve robustness in highly imbalanced medical imaging tasks without increasing architectural complexity.

Future work may explore stronger data augmentation or synthetic data generation to further enhance minority-class diversity, as well as self-supervised or domain-specific pretraining on large-scale unlabeled dermoscopic data. Evaluation on external datasets is also necessary to assess generalization and clinical applicability. Overall, this work highlights the importance of explicitly addressing class imbalance and provides a practical framework for improving automated skin lesion classification in clinically realistic settings.

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